

# SESSION 1

## 1. MySQL Workbench setup

Install MySQL Server and MySQL Workbench, then open Workbench and connect to Local instance. The Schemas panel normally shows databases such as information\_schema, mysql, performance\_schema, and sys. Take your own screenshot after connecting.

Connection: MySQL Connections > Local instance > enter root password.

## 2. Create InstaClone and Users

A database contains related tables. A table is made of columns (fields) and rows (records). PRIMARY KEY makes each user\_id unique.

```
CREATE DATABASE InstaClone;
USE InstaClone;
CREATE TABLE Users (
  user_id INT PRIMARY KEY AUTO_INCREMENT,
  username VARCHAR(50) NOT NULL,
  email VARCHAR(120) NOT NULL UNIQUE,
  followers_count INT NOT NULL DEFAULT 0
);
```

## 3. Insert sample users

INSERT adds rows. Listing column names makes the statement easier to maintain.

```
INSERT INTO Users (username,email,followers_count) VALUES
('travelwithriya','riya@example.com',1250),
('foodie_aarav','aarav@example.com',3480),
('dailyart_maya','maya@example.com',890);
SELECT * FROM Users;
```

## 4. Create Posts and relationship

Posts.user\_id is a foreign key to Users.user\_id. It prevents a post from referring to a missing user.

```
CREATE TABLE Posts (
  post_id INT PRIMARY KEY AUTO_INCREMENT,
  user_id INT NOT NULL,
  caption VARCHAR(500),
  post_date DATE NOT NULL,
  FOREIGN KEY (user_id) REFERENCES Users(user_id)
);
```

## 5. DBMS comparison

**MySQL:** popular open-source relational database for web applications, especially PHP/LAMP stacks. **PostgreSQL:** advanced open-source database used for complex queries, GIS, and data-heavy web apps. **Oracle:** commercial enterprise database used by large banks, government, and high-availability systems. **SQLite:** lightweight file-based database, common in mobile apps, local tools, and prototypes.